



Department of Mathematics, Physics and Computer Science
College of Science and Mathematics

Evolution of Websites

Presented by
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Learning Objectives

At the end of the lesson, all of you should be able to:

1. Differentiate between static and dynamic web content
2. Describe the limitations of early web technologies.
3. Identify and define the core technologies used in web design

Introduction

“The web is constantly evolving, and though it is still an infant relative to other media such as print, TV and radio, it has already amassed a rich history.”

– William Craig

Key Stages in the Evolution of Websites

World Wide Web

The WorldWideWeb (W3) is a wide-area [hypermedia](#) information retrieval initiative aiming to give universal access to a large universe of documents.

Everything there is online about W3 is linked directly or indirectly to this document, including an [executive summary](#) of the project, [Mailing lists](#) , [Policy](#) , November's [W3 news](#) , [Frequently Asked Questions](#) .

[What's out there?](#)

Pointers to the world's online information, [subjects](#) , [W3 servers](#), etc.

[Help](#)

on the browser you are using

[Software Products](#)

A list of W3 project components and their current state. (e.g. [Line Mode](#) ,X11 [Viola](#) , [NeXTStep](#) , [Servers](#) , [Tools](#) ,[Mail robot](#) ,[Library](#) .)

[Technical](#)

Details of protocols, formats, program internals etc

[Bibliography](#)

Paper documentation on W3 and references.

[People](#)

A list of some people involved in the project.

[History](#)

A summary of the history of the project.

[How can I help ?](#)

If you would like to support the web..

[Getting code](#)

Getting the code by [anonymous FTP](#) , etc.

1. Static HTML Websites (Early 1990s)

- Static HTML websites are the earliest form of websites, consisting mainly of fixed, text-based pages written manually using HTML.
- Key Characteristics:
 - Manually coded using HTML
 - Primarily text-based
 - Fixed or static content
 - Limited interactivity
 - Basic formatting such as fonts, colors, and headings
 - Each page had to be edited individually

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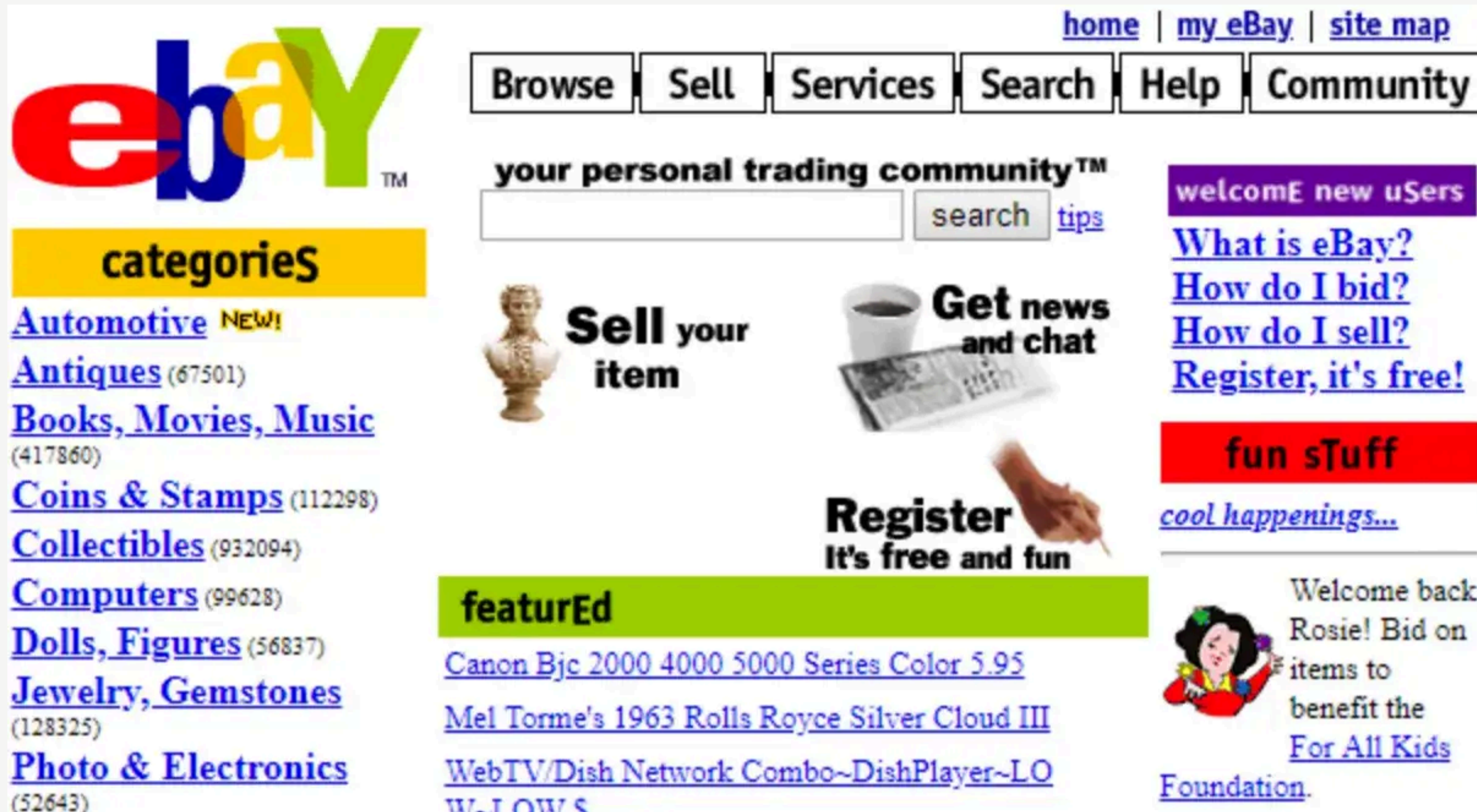
A summary of the history of the project.

[How can I help ?](#)

If you would like to support the web..

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Getting the code by [anonymous FTP](#) , etc.



The screenshot shows the eBay homepage layout. At the top right are links for [home](#), [my eBay](#), and [site map](#). Below these is a horizontal navigation bar with buttons for **Browse**, **Sell**, **Services**, **Search**, **Help**, and **Community**. The main heading is **your personal trading community™**, followed by a search input field, a **search** button, and a [tips](#) link. On the left, a yellow **categories** banner lists various product categories with their item counts: [Automotive](#) (NEW!), [Antiques](#) (67501), [Books, Movies, Music](#) (417860), [Coins & Stamps](#) (112298), [Collectibles](#) (932094), [Computers](#) (99628), [Dolls, Figures](#) (56837), [Jewelry, Gemstones](#) (128325), and [Photo & Electronics](#) (52643). In the center, there are three promotional banners: **Sell your item** with an image of a bust, **Get news and chat** with an image of a cup and newspaper, and **Register It's free and fun** with an image of a hand pointing. On the right, a purple banner says **welcome new users** with links for [What is eBay?](#), [How do I bid?](#), [How do I sell?](#), and [Register, it's free!](#). Below this is a red **fun stuff** banner with a link to [cool happenings...](#). At the bottom right, a cartoon character of Rosie the Riveter is shown next to a message welcoming back a user named Rosie to bid on items for the [For All Kids Foundation](#). A green **featurEd** banner at the bottom left lists featured items: [Canon Bjc 2000 4000 5000 Series Color 5.95](#), [Mel Torme's 1963 Rolls Royce Silver Cloud III](#), and [WebTV/Dish Network Combo~DishPlayer~LO](#).

[home](#) | [my eBay](#) | [site map](#)

Browse **Sell** **Services** **Search** **Help** **Community**

your personal trading community™

search [tips](#)

categories

[Automotive](#) **NEW!**

[Antiques](#) (67501)

[Books, Movies, Music](#) (417860)

[Coins & Stamps](#) (112298)

[Collectibles](#) (932094)

[Computers](#) (99628)

[Dolls, Figures](#) (56837)

[Jewelry, Gemstones](#) (128325)

[Photo & Electronics](#) (52643)

Sell your item

Get news and chat

Register
It's free and fun

welcome new users

[What is eBay?](#)

[How do I bid?](#)

[How do I sell?](#)

[Register, it's free!](#)

fun stuff

[cool happenings...](#)

Welcome back Rosie! Bid on items to benefit the [For All Kids Foundation](#).

featurEd

[Canon Bjc 2000 4000 5000 Series Color 5.95](#)

[Mel Torme's 1963 Rolls Royce Silver Cloud III](#)

[WebTV/Dish Network Combo~DishPlayer~LO](#)

2. Introduction of Graphics (Mid-1990s)

- Images and graphical elements were added to improve the appearance of websites and make information more attractive to users.
- Key Characteristics:
 - Images were added to webpages
 - GIF images became widely used
 - Websites became more visually appealing
 - Graphics were still relatively limited because of slow Internet connections

Yahoo! (1996)



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W3C WORLD WIDE WEB
CONSORTIUM

Leading the Web to its Full Potential...

The World Wide Web Consortium Issues HTML 4.0 as a W3C Recommendation

"Insisting on HTML 4.0 compliance now will preserve your free choice of suppliers of Web software, tools and applications well into the future."
— Tim Berners-Lee, W3C Director and inventor of the World Wide Web

W3C Issues XML 1.0 as a Proposed Recommendation

Register for the 7th International World Wide Web Conference (WWW7) April 14-18, 1998

About W3C
who we are, why we're here, and how to become a Member

About the Web
history and general information about the Web

Press Information
news items, news releases, W3C in the press

People of the W3C
about us — with a personal note

How to Contact Us
we're global — North America, Europe and Asia

News & Announcements
keep up with the latest events at the Consortium

Member Area
for W3C Members only

Technical Areas
the "big picture" of what we do

Technical Reports & Publications
current Recommendations, Working Drafts, Notes and Submissions

Sample Code
W3C Reference Library, toolset and experimental software

User Interface

[HTML](#)
[Style Sheets](#)
[Document Object Model](#)
[Math](#)
[Graphics](#)
[Internationalization](#)
[Forms](#)
[Access](#)

Technology and Society

[Digital Signature Initiative](#)
[Metadata](#)
[PDR](#)
[Privacy \(P4\)](#)
[Interest Groups](#)

Architecture

[HTTP](#)
[HTTP/NG](#)
[Recommended Multimedia](#)
[XML](#)
[Javas](#)

Web Accessibility Initiative

W3C Services

[HTML Validation Service](#)
[Making Links](#)

Historical

Moved

3. Table-Based Layouts (Late 1990s - Early 2000s)

- Websites started using tables for layout, allowing for more structured designs.
- CSS was introduced but not widely adopted for layout purposes.
- Key Characteristics:
 - Tables were used to structure webpage layouts
 - Allowed developers to create more organized designs
 - Supported more complex page arrangements
 - WYSIWYG tools such as Dreamweaver became popular

Yahoo! (2002)



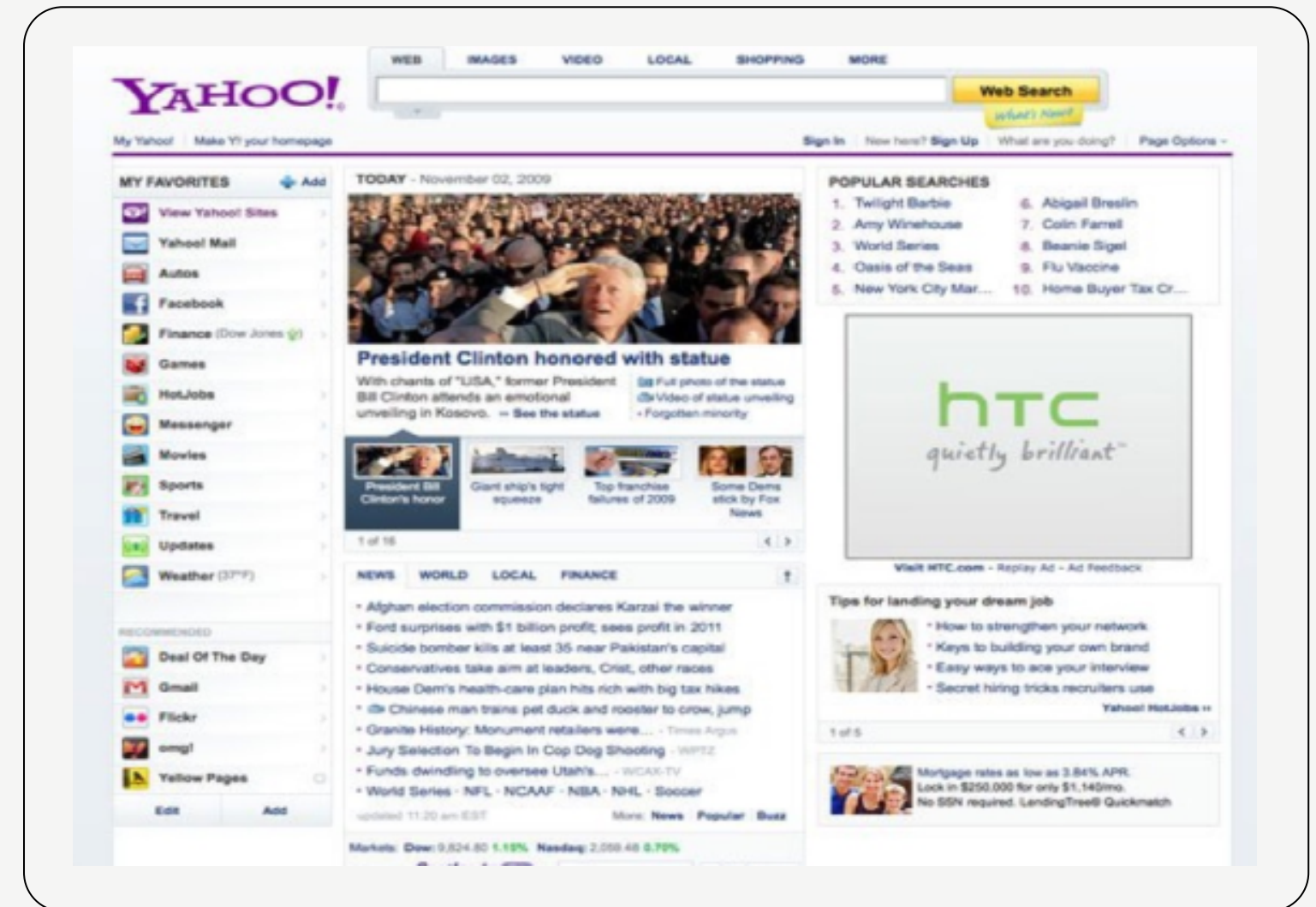
4. Flash and Multimedia (Late 1990s - Early 2000s)

- Flash technology (1996) allowed for interactive multimedia content and animations.
- Many websites used Flash intros, but these were often criticized for being slow to load and inaccessible.
- Interactive web pages that allow users to not only read static content, but also to interact with web content, began.



5. CSS-Based Layouts and Web Standards (Early 2000s)

- CSS-based designs started gaining in popularity after the dotcom boom in the early 2000's.
- Introduced a more organized approach to website development by separating the presentation or appearance of a webpage from its content and structure
- Consistent design across multiple pages.
- Smaller and more efficient webpages.



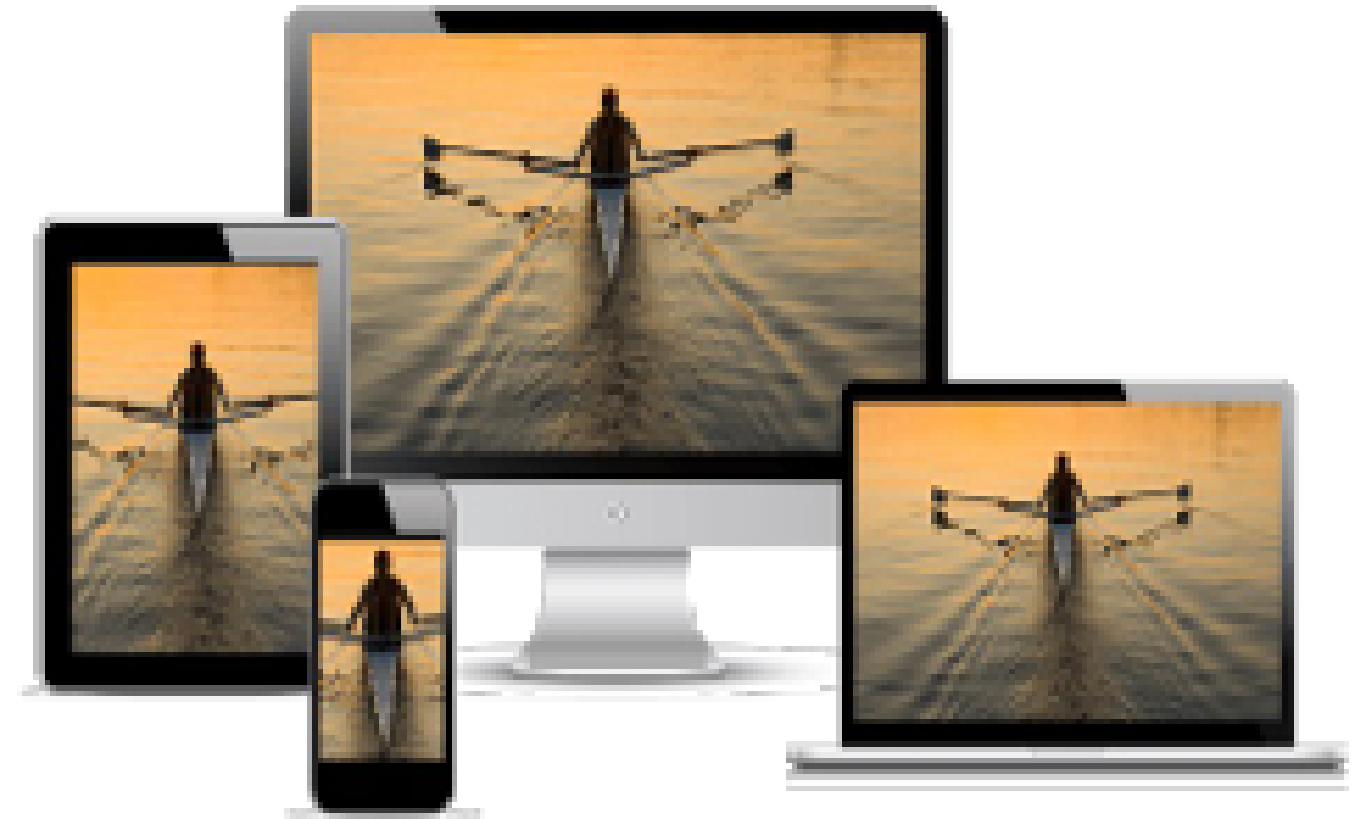
6. Web 2 & User-Centric Design (Mid-2000s)

- Web 2.0 refers to the development of websites that became more interactive, dynamic, collaborative, and centered on user participation.
- Several platforms and websites exemplify the Web 2.0 era by embracing and promoting user-generated content.
- Key Characteristics:
 - User-generated content
 - Social networking
 - Online collaboration
 - Increased sharing and exchange of data

Social Media Platform	Data Produced per Day (June 2024)
Facebook	4,000 terabytes (TB)
YouTube	4.3 petabytes (PB)
Twitter	560 gigabytes (GB)
Instagram	100 TB (estimated)
TikTok	7.35 TB (estimated)
Snapchat	30 TB (estimated)
WhatsApp	10 TB (estimated)
Telegram	5 TB (estimated)

7. Responsive Web Design (2010s)

- This means layouts and content can change based on device characteristics
- The proliferation of mobile devices led to the need for responsive web design.
- CSS media queries and flexible layouts allowed websites to adapt to various screen sizes.

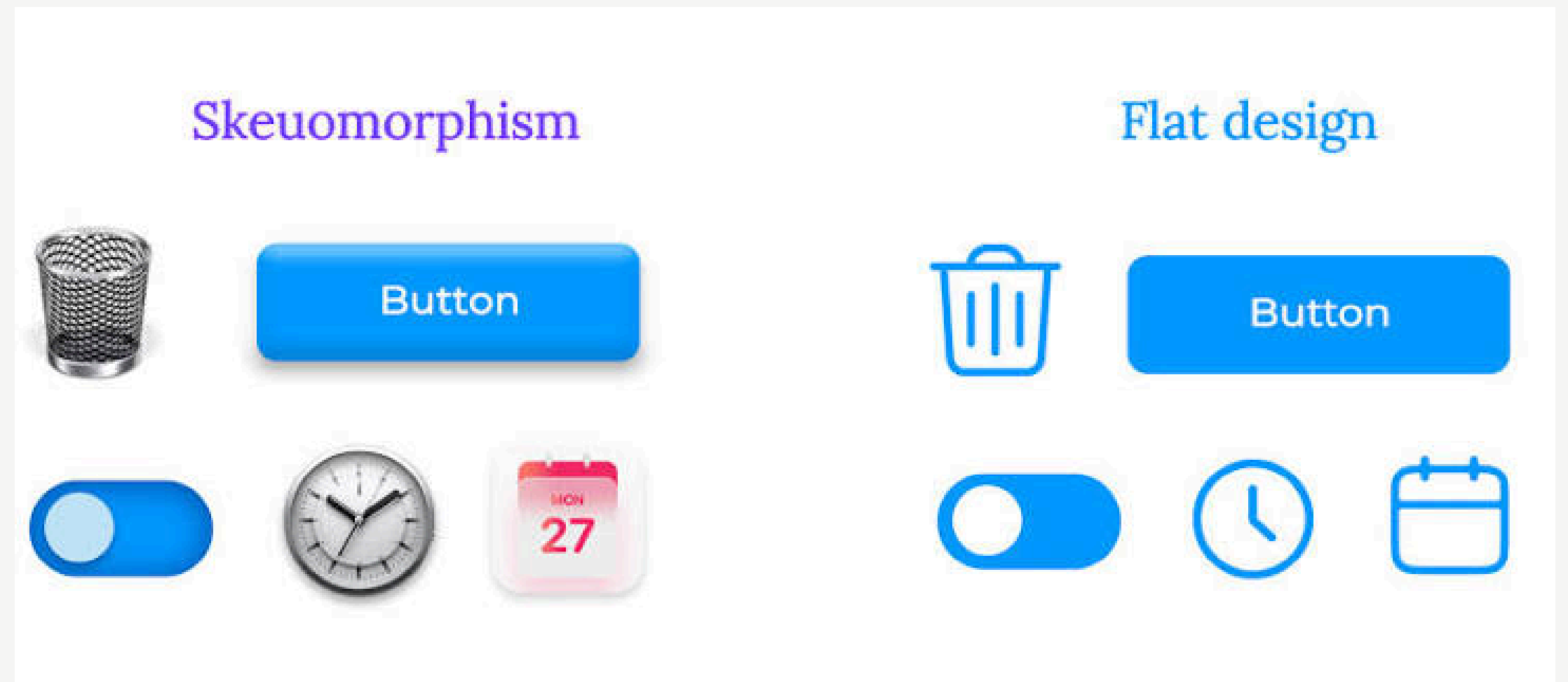
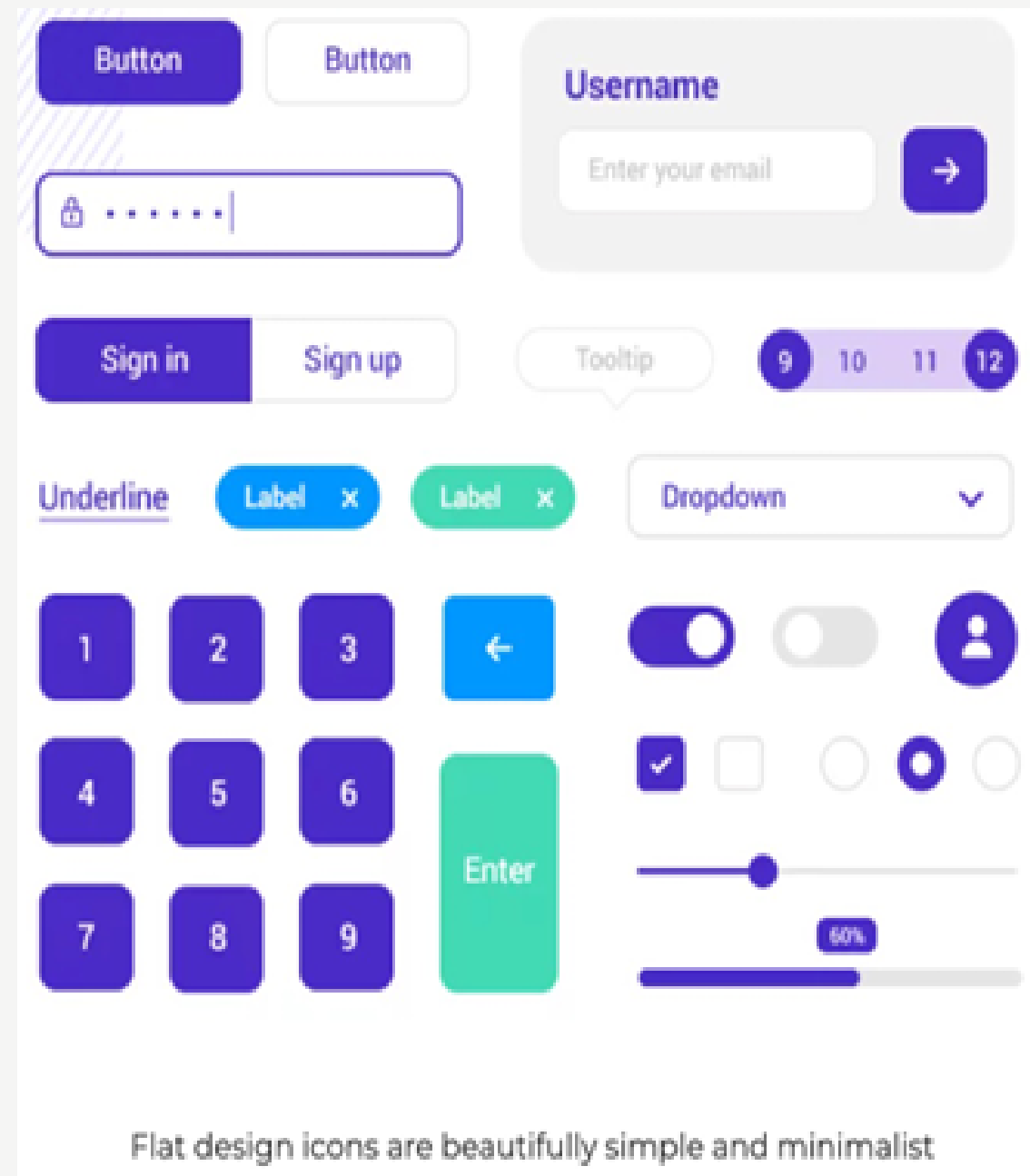


8. Flat and Material Design (Mid-2010s)

- Flat design and Material Design are two popular design philosophies in the field of user interface (UI) and user experience (UX) design.

Flat Design

- Minimalist interface design approach that uses simple shapes, two-dimensional visuals, clear typography, and limited decorative effects to make interfaces clean, readable, and easy to use.



Advantages

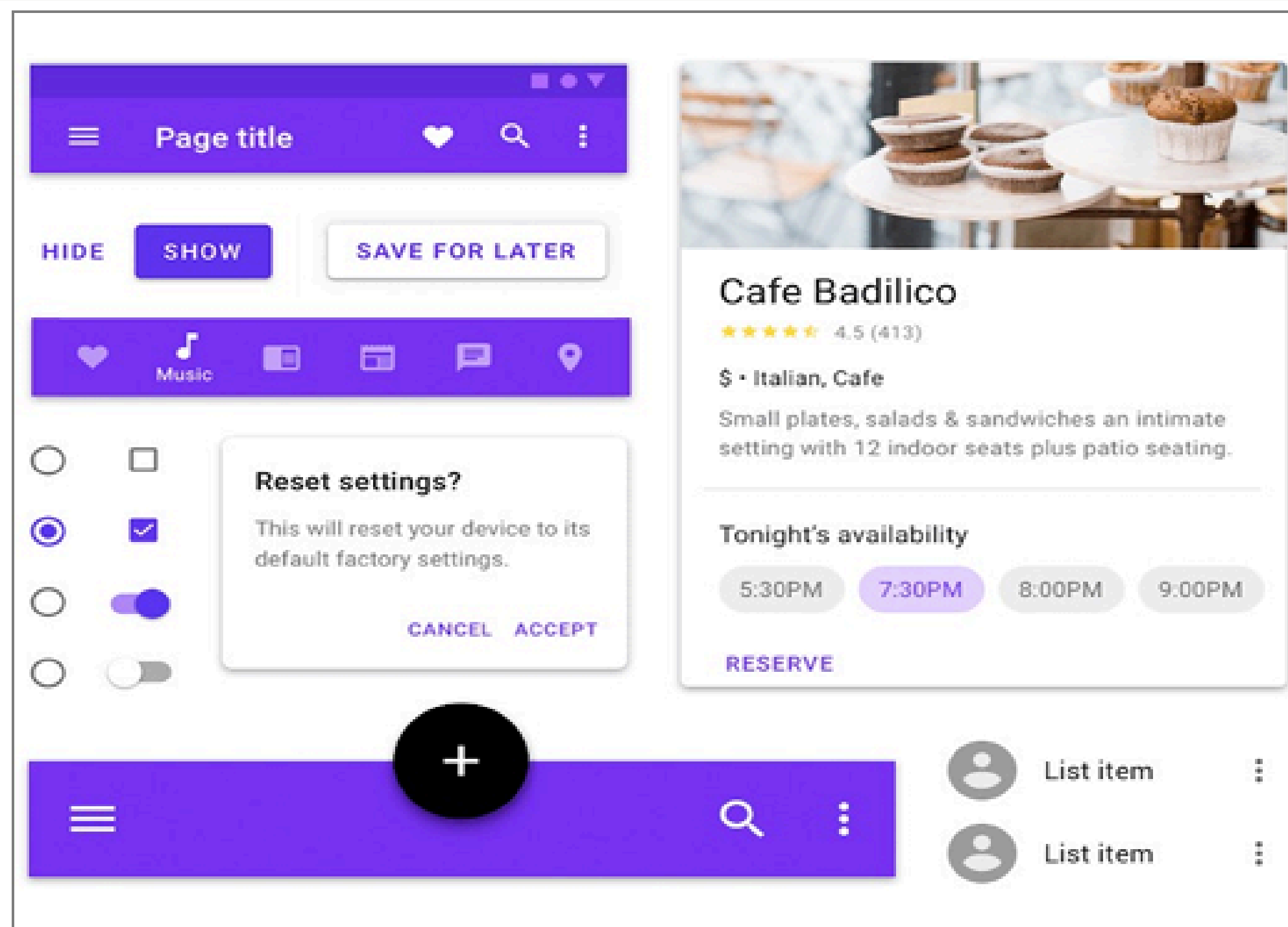
- Uncluttered, distraction-free design
- High readability with clear typography
- Easily adjusted for responsive design
- Faster to load in browsers and apps

Disadvantages

- Lack of depth makes elements look less clickable
- Creative design and standing out is harder
- No design guidelines like other design systems

Material Design

- Material Design is a design system developed by Google that provides guidelines for creating consistent and intuitive user interfaces across different devices and platforms.
- It uses concepts such as depth, shadows, movement, and layering to help users understand how interface elements behave.



Material Design's elements mimic paper in space

Advantages

- Unified and simplistic interface
- Principles and goals provide consistency for designers
- Z-axis creates depth
- Intuitive to use
- Uses motion to show users what is happening on screen

Disadvantages

- Floating action buttons can sometimes be unnecessary.
- Designs only valid for Android devices.
- Heavily associated with Google with less room for branding.
- UIs designed without motion often lack intuitiveness.

9. Single-Page Applications (SPAs) and Progressive Web Apps (PWAs) (Late 2010s - Present)

Single-Page Applications (SPAs)

- type of web application/ website that loads and operates on a single HTML page by using JavaScript framework to load content dynamically and overwrites the existing page with new data from a web server rather than reloading pages individually from scratch.
- Page content updates in real time based on user actions with quick transitions and without refreshing.

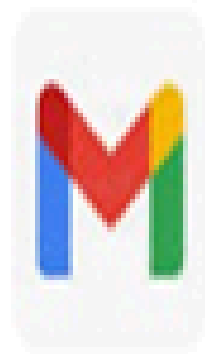
Characteristics of SPAs

1. **Dynamic Content** - SPAs load data from the server asynchronously (typically using AJAX requests) and update the page's content without requiring a full page reload.
2. **Client-Side Routing** - SPAs often use client-side routing to manage different "views" or sections within the application.
3. **Faster Interactions** - fetch the data needed for a specific interaction.
4. **Challenges with SEO** - search engines traditionally rely on crawling static HTML content, which SPAs generate dynamically.

Advantages of SPAs

- Improved performance
- Easy to build
- Offline functionality

From sources across the web



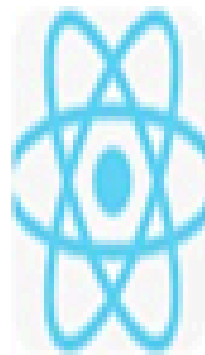
Gmail



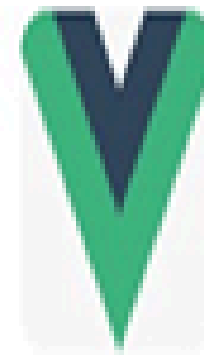
Facebook



Netflix



React



Vue.js



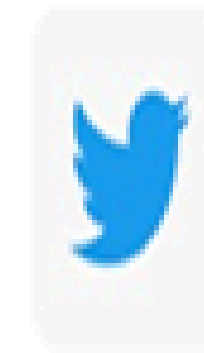
Angular



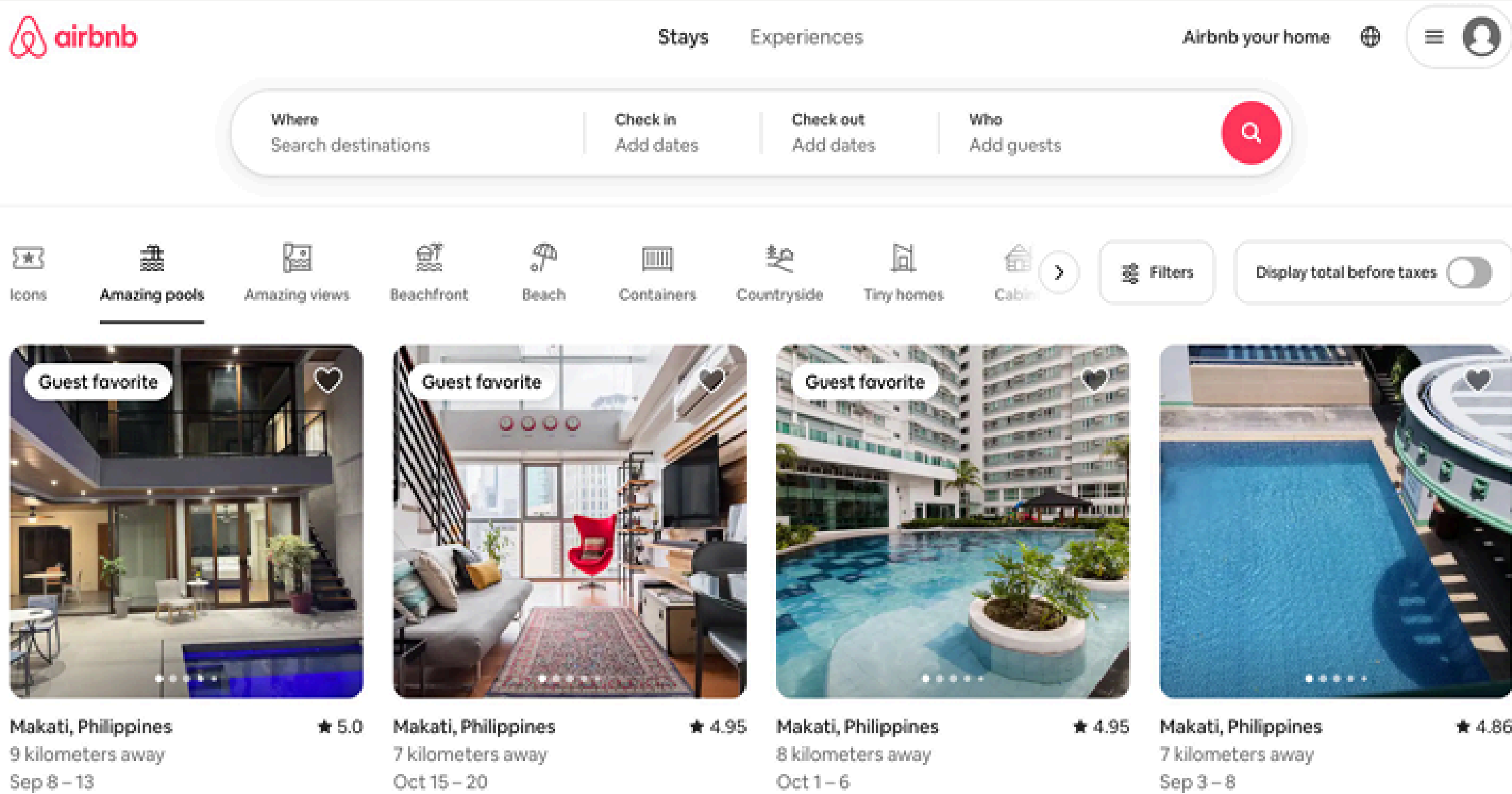
Airbnb



PayPal



Twitter



Progressive Web Apps (PWAs)

- are a type of web application that combines the best features of both websites and native mobile apps.
- The concept was introduced in 2015 by Google to enable businesses to offer improved user experience across devices using a single codebase.
- PWAs are designed to work on any device, including smartphones, tablets, and desktops, and can be accessed through a web browser.
- They're designed to be reliable, fast, and engaging, and offer features such as offline support and push notifications.

Progressive Web Apps (PWAs)

- PWAs are built using standard web technologies such as HTML, CSS, and JavaScript

Characteristics of PWAs

1. **Offline Capabilities** - PWAs can work offline or in low network conditions by caching essential assets and data.
2. **Responsive Design** - PWAs are designed to be responsive and work well on various devices and screen sizes, from smartphones to desktops

Characteristics of PWAs

3. **App-Like Interactions** - PWAs provide a more app-like experience by offering smooth animations, gestures, and transitions.
4. **Push Notifications** - PWAs can send push notifications to users, even when the app is not open, to engage users and provide updates.
5. **Installation** - Users can "install" PWAs on their device's home screen, which creates an icon and provides quick access to the web app, making it feel more like a native app.
6. **Security** - PWAs must be served over HTTPS to ensure data security.

Advantages of PWAs

- Fast and reliable performance
- Cost effective to build
- Offline functionality
- Automatic update
- Quick installation

Feature	Web App	Native App
Development	Uses web technologies (HTML, CSS, JS)	Requires platform-specific languages and tools (e.g., Swift for iOS, Kotlin for Android)
Platform compatibility	Cross-platform (works on various devices)	Platform-specific (designed for a particular OS)
Performance	Generally slower than native apps	Often faster and more responsive
Offline functionality	Limited or non-existent	Can be built-in
Device features	Limited access to device-specific features (camera, GPS)	Full access to device-specific features
User experience	Can be less immersive than native apps	Typically offers a more immersive and tailored experience
Deployment	Easier and faster (no app store approval)	Requires app store approval, which can be time-consuming
Cost	Generally lower development and maintenance costs	Higher development and maintenance costs
Updates	Easier to update (pushed directly to users)	Require app store updates, which can take time

SPA

- less secure
- web applications that are designed to be fast and responsive
- slower than PWAs when it comes to initial load time

PWA

- they are discoverable and crawlable by search engines
- more expensive
- often easier to maintain and update

E-commerce and Retail

- AliExpress - Offers a PWA for a faster and more app-like shopping experience.
- Starbucks - Provides a PWA for mobile ordering and payment.

News and Media

- The Washington Post - Offers a PWA for a more immersive news experience.
- The New York Times - Provides a PWA for offline reading and push notifications.

Social Media

- Pinterest - Offers a PWA for faster browsing and offline pinning.
- Instagram - Provides a PWA for web-based access to their platform.

Other Examples

- Google Maps - Offers a PWA for offline navigation.
- Trivago - Offers a PWA for hotel searches

10. Voice Interfaces and AI Integration (Present and Future)

- Voice search and AI-powered chatbots are becoming integral parts of web design.
- Websites are being designed to interact with users through voice commands and natural language processing.

Thank you!